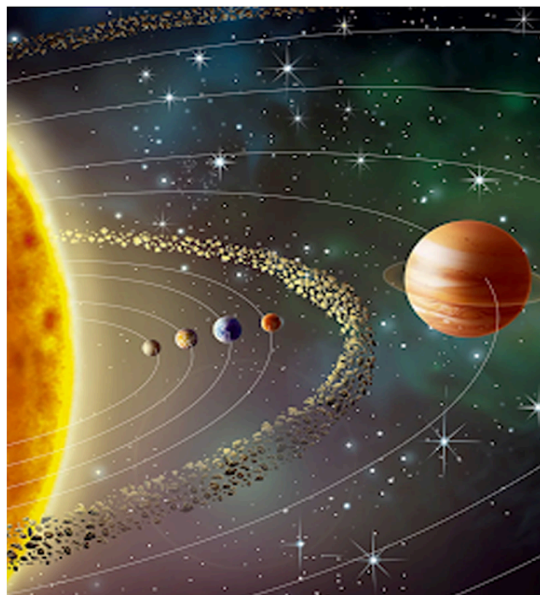
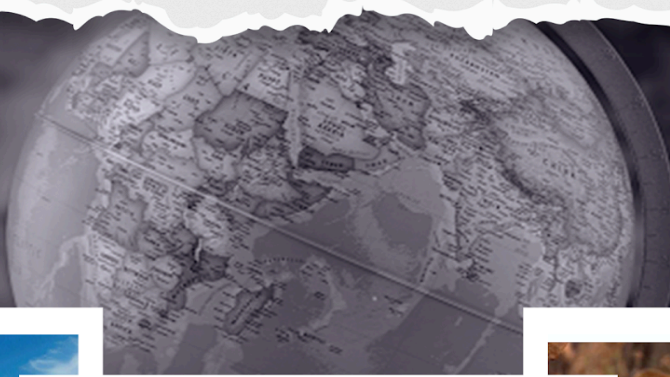


General Knowledge

World Geography

STUDY NOTES

Geomorphology: Earthquake

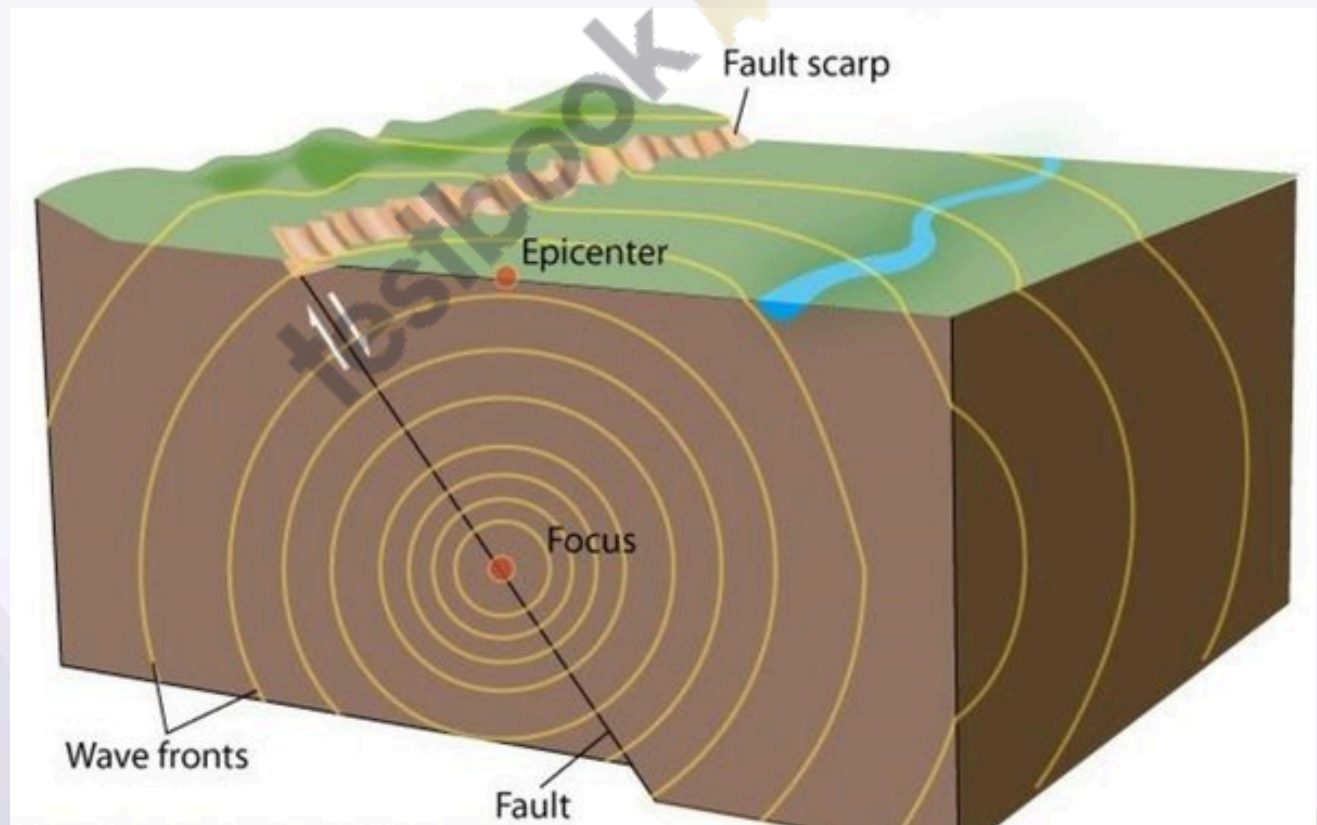


Geomorphology: Earthquake

An **earthquake** is the shaking of the earth's surface, caused by the seismic waves or **earthquake waves** that are generated due to a sudden movement (sudden release of energy) in the earth's crust (shallow-focus earthquakes) or upper mantle (some shallow-focus and all intermediate and deep-focus earthquakes). A **seismograph** is an instrument used to detect and record earthquakes.

Focus and epicentre

- The point where the energy is released is called the **focus** or the **hypocentre** of an earthquake.
- The point on the surface directly above the focus is called the **epicentre**, it is the first surface point to experience earthquake waves.



Foreshocks and aftershocks

- A major or even moderate earthquake is usually followed by many lesser-size earthquakes known as aftershocks.
- A mild earthquake preceding the violent shaking movement of an earthquake is known as a foreshock.

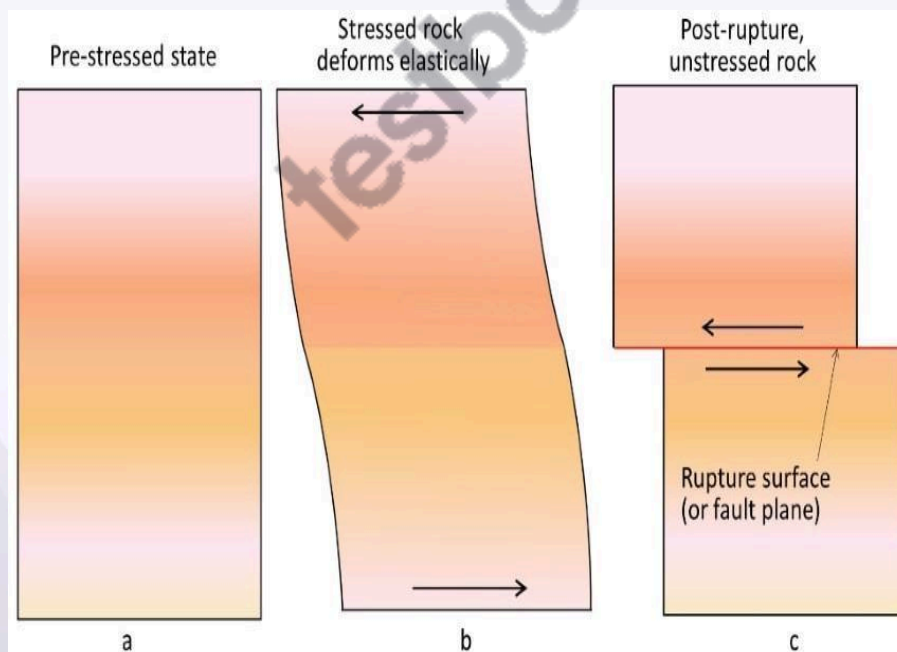
Swarms

- Large numbers of small earthquakes may occur in a region for months without a major earthquake. Such a series of earthquakes are called earthquake swarms.

Causes of Earthquakes

Fault Zones -

- The immediate cause of shallow earthquakes is the sudden release of stress along a fault crack in the earth's crust.
- Sudden slipping of rock formations along faults in the earth's crust happens due to the constant change in volume and density of rocks due to intense temperature and pressure in the earth's interior.



- The longer the length and the wider the width of the faulted area, the larger the resulting magnitude.

Volcanic activity -

- Volcanic activity also can cause an earthquake, **but the earthquakes of volcanic origin are generally less severe and more limited in extent** than those caused by fracturing of the earth's crust.
- **Earthquakes in volcanic regions are caused by the consequent release of elastic strain energy both by tectonic faults and the movement of magma in volcanoes.**
- There is a clear correspondence between the geographic distribution of volcanoes and major earthquakes, particularly in the Circum-Pacific Belt and along oceanic ridges.

Human Induced Earthquakes -

- Human Human-induced earthquakes refers to typically minor earthquakes and tremors that are **caused by human activity like mining, large-scale petroleum extraction, artificial lakes, nuclear tests etc.**

Reservoir-induced seismicity -

- **The pressure offered by a column of water in a large and deep artificial lake alters stresses along an existing fault or fracture.** Also, the percolation of water weakens the soil structure and lubricates the faults.
- Loading and unloading of water can significantly change the stress. This significant change in stress can lead to a sudden movement along the fault or fracture, resulting in an earthquake.
- The 6.3 magnitude 1967 **Koynanagar earthquake** occurred near the Koyna Dam reservoir in Maharashtra and claimed more than 150 lives. There have been several earthquakes of smaller magnitude since then.

Earthquakes based on the depth of focus

- For scientific purposes, this earthquake depth range of 0 - 700 km is divided into three zones: **shallow, intermediate, and deep.**
- Shallow-focus earthquakes are found within the earth's outer crustal layer, while deep-focus earthquakes occur within the deeper subduction zones of the earth.

- **Shallow earthquakes are 0 – 70 km deep.**
- **Intermediate earthquakes are 70 – 300 km deep.**
- **Deep earthquakes are 300 – 700 km deep.**
- Of the total energy released in earthquakes, about 12-15 per cent comes from intermediate earthquakes, about 3-5 per cent from deeper earthquakes and about 70-85 per cent from shallow earthquakes.
- A quake's destructive force depends not only on the energy released but also on location, distance from the epicentre and depth.
- On 24 August 2016, a 6.2 earthquake rocked Central Italy killing about 300 people. An even bigger 6.8 hit Myanmar the same day killing just a few people.
- Italy's quake was very shallow, originating within 10 kilometres underground. By contrast, the quake in Myanmar was deeper — 84 kilometres.

Seismic wave

- **Seismic waves are waves of energy that travel through the Earth's layers and are a result of earthquakes, volcanic eruptions, magma movement, large landslides and large human-made explosions.**
- The terms seismic waves and earthquake waves are often used interchangeably.

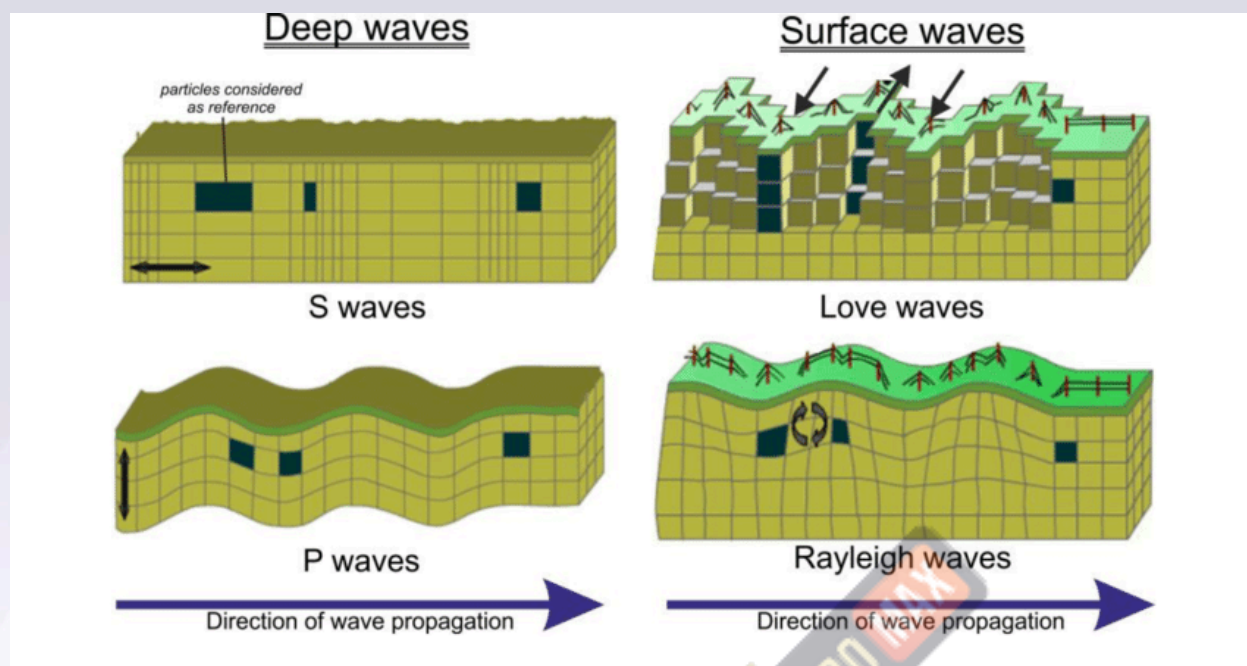
How are earthquake waves produced?

- **The abrupt release of energy along a fault (sharp break in the crustal layer) causes earthquake waves.**
- Rock layers along a fault tend to move in opposite directions due to the force exerted on them but are held in place by counteracting frictional force exerted by the overlying rock strata.
- **The pressure on the rock layers builds up over a period and overcomes the frictional force resulting in a sudden movement** generating shockwaves (seismic waves) that travel in all directions.

Types of Seismic waves or earthquake waves

The seismic waves or earthquake waves are basically of two types —

- 1) **Body Waves**
- 2) **Surface Waves**



Body Waves:

- Body waves are generated due to the release of energy at the focus and move in all directions travelling through the interior of the earth. Hence, the name body waves.
- There are two types of body waves: The **P-waves** or primary waves and the **S-waves** or secondary waves.

Primary Waves (P-waves)

- Primary waves are called so because they are the **fastest** among the seismic waves and hence are recorded first on the seismograph.
- P-waves are -
- **Longitudinal waves** because the displacement of the medium is in the same direction as the direction of propagation of the wave;
- **Compressional waves** because they produce compression and rarefaction when travelling through a medium; or
- **Pressure waves** because they produce increases and decreases in pressure in the medium.
- These waves are of relatively high frequency and are the least destructive among the earthquake waves.

- They can travel in all mediums, and their velocity depends on shear strength (elasticity) of the medium. Hence, the velocity of the P-waves in Solids > Liquids > Gases. P-wave velocity in earthquakes is in the range 5 to 8 km/s.

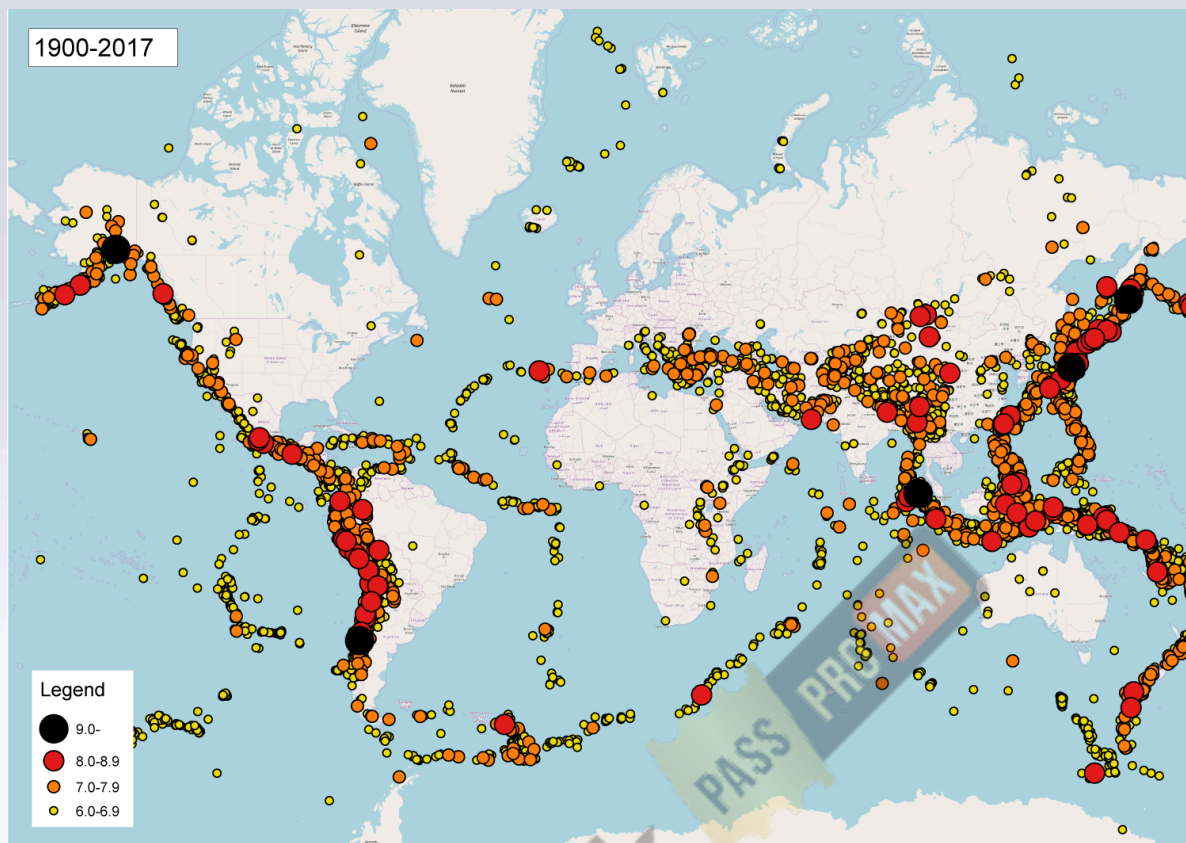
Secondary Waves (S-waves)

- Secondary waves recorded second on the seismograph are also **called transverse waves or shear waves or distortional waves**.
- They are analogous to water ripples or light waves.
- **Transverse waves or shear waves mean that the direction of vibrations of the particles in the medium is perpendicular to the direction of propagation of the wave.** Hence, they create troughs and crests in the material through which they pass (they distort the medium).
- **S-waves arrive at the surface after the P-waves.**
- These waves are of high frequency and possess slightly higher destructive power compared to P-waves.
- S-waves cannot pass through fluids (liquids and gases) as fluids do not support shear stresses.

Surface waves (L-Waves):

- **The body waves interact with the surface rocks and generate new sets of waves called surface waves (long or L-waves).** These waves move only along the surface.
- They are low-frequency transverse waves and they develop in the immediate neighbourhood of the epicentre and affect only the surface of the earth and die out at smaller depth.
- They lose energy more slowly with distance than the body waves because they travel only across the surface unlike the body waves which travel in all directions.
- Particle motion of surface waves (amplitude) is larger than that of body waves, so surface waves are the most destructive among the earthquake waves.
- They are slowest among the earthquake waves and are recorded last on the seismograph.

Distribution of Earthquakes



- **Earth's major earthquakes occur mainly in belts coinciding with the margins of tectonic plates.** The most important earthquake belt is the Circum-Pacific Belt, which affects many populated coastal regions around the Pacific Ocean—for example, those of New Zealand, New Guinea, Japan, the Aleutian Islands, Alaska, and the western coasts of North and South America.
- Because at many places the Circum-Pacific Belt is associated with volcanic activity, it has been popularly dubbed the “**Pacific Ring of Fire.**”
- **The Pacific Ring of Fire accounts for about 68 per cent of all earthquakes.**
- **A second belt, known as the Alpine Belt (Himalayas and Alps).** The energy released in earthquakes from this belt is about 15 percent of the world total.
- **The mid-world mountain belt (Alpine Belt)** extends parallel to the equator from Mexico across the Atlantic Ocean, the Mediterranean Sea from Alpine-Caucasus ranges to the Caspian, Himalayan mountains and the adjoining lands.
- There also are striking connected belts of seismic activity, mainly along oceanic ridges—including those in the Arctic Ocean, the Atlantic Ocean, and the western Indian Ocean—and along the rift valleys of East Africa.

Richter magnitude scale

- Charles F. Richter developed the Richter magnitude scale (M_L) for measuring the strength (amount of energy released) of earthquakes in the 1930s.
- Under the Richter magnitude scale, an increase of one step corresponds to about a 32 times increase in the amount of energy released, and an increase of two steps corresponds to a 1,000 times increase in energy.
- Thus, an earthquake of M_w of 7.0 releases about 32 times as much energy as one of 6.0 and nearly 1,000 times ($\sim 32 \times 32$) one of 5.0.
- Richter scale is only effective for regional earthquakes no greater than M_5 . Moment magnitude scale is more effective for large earthquakes.

Magnitude	Description	Average earthquake effects	Frequency of occurrence
1.0–1.9	Micro	• Micro earthquakes, not felt, or felt rarely. • They are recorded by seismographs.	Several million per year
2.0–2.9	Minor	• Felt slightly by some people. • No damage to buildings.	Over one million per year
3.0–3.9		• Often felt by people, but very rarely causes damage. • Shaking of indoor objects can be noticeable.	Over 100,000 per year
4.0–4.9	Light	• Noticeable shaking of indoor objects. • They are felt by most people in the affected area. • Slightly felt outside. • Generally, causes none to minimal damage. • Some objects may fall off shelves or be knocked over.	10,000 to 15,000 per year

5.0–5.9	Moderate	• Can cause damage of varying severity to poorly constructed buildings. • At most, none to slight damage to all other buildings. • Felt by everyone.	1,000 to 1,500 per year
6.0–6.9	Strong	• Damage to a moderate number of well-built structures in populated areas. • Earthquake-resistant structures survive with slight to moderate damage. • Poorly designed structures receive moderate to severe damage. • Felt in wider areas; up to hundreds of kilometres from the epicentre. • Strong to violent shaking in the epicentral area.	100 to 150 per year • 2011 Christchurch earthquake (6.2)
7.0–7.9	Major	• Causes damage to most buildings, some to partially or completely collapse or receive severe damage. Well-designed structures are likely to receive damage. Felt across great distances with major damage mostly limited to 250 km from epicentre.	10 to 20 per year • 1819 Rann of Kutch earthquake (7.7–8.2) • 2001 Gujarat earthquake (7.7)

8.0–8.9	Great	- Major damage to buildings, structures likely to be destroyed. - Will cause moderate to heavy damage to sturdy or earthquake-resistant buildings. - Damage in large areas. - Felt in extremely large regions.	One per year 1556 Shaanxi earthquake (8.0) 1950 Assam–Tibet earthquake (8.6) 2008 Sichuan earthquake (8.0) 2010 Chile earthquake (8.8)
9.0 and greater		- At or near total destruction – severe damage or collapse to all buildings. - Heavy damage and shaking extends to distant locations. - Permanent changes in ground topography.	One per 10 to 50 years 1960 Valdivia earthquake, Chile (9.4–9.6) 1964 Alaska earthquake (9.2) 2004 Indian Ocean earthquake (9.1–9.3) 2011 Tōhoku earthquake, Japan (9.1)

Earthquake zones of India -

- The latest seismic zone map prepared by the **National Disaster Management Authority** reveals that nearly **59% of India's land area** is prone to moderate or severe earthquakes.
- This earthquake zoning map divides India into **five different zones of earthquake** intensity and highlights the location that falls under them.

